FOCUS 2 • PHYSICAL CHEMISTRY • 物理化学

The First Law (continued)

热力学第一定律（续）

Topics 2C Thermochemistry · 2D State functions · 2E Adiabatic changes

WHERE WE ARE

Finishing the First Law

- 2A (internal energy) and 2B (enthalpy) are done.

Property	Equation	Comment	Equation number
First Law of thermodynamics	$\Delta U = q + w$	Convention	2A.2
Work of expansion	$dw = -p_{ex}dV$		2A.5a
Work of expansion against a constant external pressure	$w = -p_{ex}\Delta V$	$p_{ex} = 0$ for free expansion	2A.6
Reversible work of expansion of a gas	$w = -nRT \ln(V_f/V_i)$	Isothermal, perfect gas	2A.9
Internal energy change	$\Delta U = q_V$	Constant volume, no other forms of work	2A.11b
Heat capacity at constant volume	$C_V = (\partial U/\partial T)_V$	Definition	2A.14

Property	Equation	Comment	Equation number
Enthalpy	$H = U + pV$	Definition	2B.1
Heat transfer at constant pressure	$dH = dq_p$, $\Delta H = q_p$	No additional work	2B.2
Relation between ΔH and ΔU at a temperature T	$\Delta H = \Delta U + \Delta n_g RT$	Molar volumes of the participating condensed phases are negligible	2B.4
Heat capacity at constant pressure	$C_p = (\partial H/\partial T)_p$	Definition	2B.5
Relation between heat capacities	$C_p - C_V = nR$	Perfect gas	2B.9

Examination content is limited to the key points presented in the PPT materials

REFERENCE · 对照表

Key concepts · 重要概念

Key concepts · 中文对照 · phonetics

thermochemistry	热化学 /ˌθɜːməʊˈkemɪstri/	**compressibility**	压缩系数 /kəmˌpresəˈbɪləti/
enthalpy H	焓 H /ˈenθəlpi/	**isenthalpic**	等焓的 /ˌaɪsenˈθælpɪk/
vaporization	蒸发 /ˌveɪpəraɪˈzeɪʃn/	**adiabatic**	绝热 /ˌædiəˈbætɪk/
transition	转化、跃迁、过渡 /trænˈzɪʃn/	**adiabat**	绝热线 /ˈædiəbæt/
fusion	熔化 /ˈfjuːʒn/		
combustion	燃烧、氧化/kəmˈbʌstʃən/		
stoichiometric	化学计量的 /ˌstɔɪkiəˈmetrɪk/		

TOPIC 2C

Thermochemistry

热化学

➤ Why do you need to know this material?

Thermochemistry is one of the principal applications of thermodynamics in chemistry. Thermochemical data provide a way of assessing the heat output of chemical reactions, including those involved with the combustion of fuels and the consumption of foods. The data are also used widely in other chemical applications of thermodynamics.

➤ What is the key idea?

Reaction enthalpies can be combined to provide data on other reactions of interest.

1. Standard reaction enthalpy · 标准反应焓
2. Hess's law · Hess 定律
3. Kirchhoff's law · Kirchhoff 定律
4. Experimental techniques · 实验技术

Standard enthalpy change

- The study of the energy transferred as heat during the course of chemical reactions is called **thermochemistry**热化学
- $\Delta H^{\ominus}$ is the enthalpy change with all species in standard states.
  标准反应焓
- The standard state of a substance at a specified temperature is its pure form at 1 bar.
  1 bar = 100 kPa ≈ 0.9869 atm. Conventional T = 298.15 K.
  标准态
- Standard enthalpy of vaporization, $\Delta_{vap}H^{\ominus}$, the enthalpy change per mole of molecules when a pure liquid at 1 bar vaporizes to a gas at 1 bar. （标准摩尔蒸发焓$\Delta_{vap}H_m^{\ominus}$）

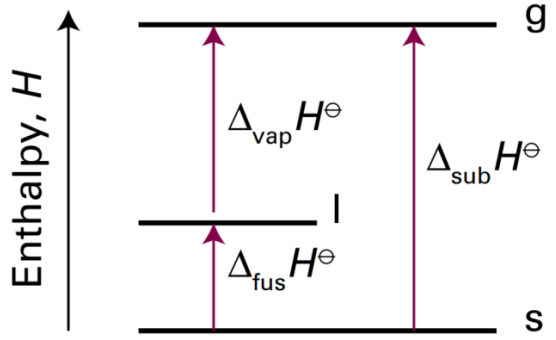
$$H_2O(l) \rightarrow H_2O(g) \qquad \Delta_{vap}H^{\ominus}(373\,K) = +40.66\,kJ\,mol^{-1}$$

· The superscript $^{\ominus}$ denotes the standard state.

Enthalpies of physical change

- The standard molar enthalpy change that accompanies a change of physical state is called the standard enthalpy of transition标准转化焓$\Delta_{trs}H^{\ominus}$ （$\Delta_{vap}H^{\ominus}$）
- Standard enthalpy of fusion, $\Delta_{fus}H^{\ominus}$, the standard molar enthalpy change accompanying the conversion of a solid to a liquid

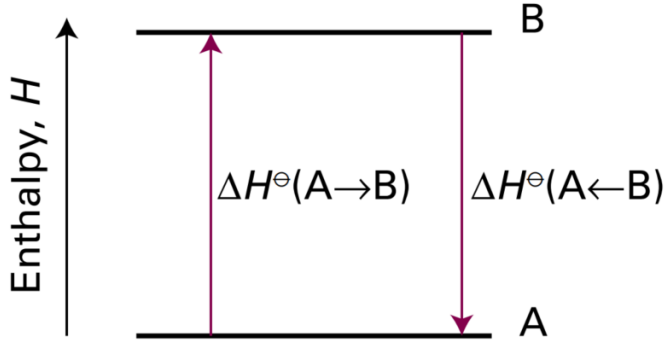
$$H_2O(s) \rightarrow H_2O(l) \qquad \Delta_{fus}H^{\ominus}(273\,K) = +6.01\,kJ\,mol^{-1}$$



$$\Delta_{sub}H^{\ominus} = \Delta_{fus}H^{\ominus} + \Delta_{vap}H^{\ominus}$$

Sublimation升华

$$\Delta H^{\ominus}(A \rightarrow B) = -\Delta H^{\ominus}(A \leftarrow B)$$



Enthalpy is a state function

Enthalpies of chemical change

- Standard reaction enthalpy标准反应焓$\Delta_r H^{\ominus}$ （标准摩尔反应焓$\Delta_r H_m^{\ominus}$，ξ=1）

$$\xi = \frac{\Delta n}{\nu} \qquad \Delta_r H_m = \frac{\Delta_r H}{\xi}$$

$$CH_4(g) + 2\,O_2(g) \rightarrow CO_2(g) + 2\,H_2O(l)\ \Delta_r H^{\ominus} = -890\,kJ\,mol^{-1}$$

- For a reaction of the form 2 A + B → 3 C + D the standard reaction enthalpy would be

$$\Delta_r H^{\ominus} = \{3H_m^{\ominus}(C) + H_m^{\ominus}(D)\} - \{2H_m^{\ominus}(A) + H_m^{\ominus}(B)\}$$

$$\Delta_r H^{\ominus} = \sum_{\text{Products}} \nu H_m^{\ominus} - \sum_{\text{Reactants}} \nu H_m^{\ominus}$$

Standard reaction enthalpy [definition] (2C.3)

Per ν_x mole X

Enthalpies of chemical change

- Standard enthalpy of combustion燃烧, $\Delta_c H^{\ominus}$, the standard reaction enthalpy for the complete oxidation of an organic compound to CO_2 gas and liquid H_2O if the compound contains C, H, and O, and to N_2 gas if N is also present.

Brief illustration 2C.1

The combustion of glucose is

$$C_6H_{12}O_6(s) + 6\,O_2(g) \rightarrow 6\,CO_2(g) + 6\,H_2O(l)$$
$$\Delta_c H^{\ominus} = -2808\,\text{kJ mol}^{-1}$$

The value quoted shows that 2808 kJ of heat is released when 1 mol $C_6H_{12}O_6$ burns under standard conditions (at 298 K). More values are given in Table 2C.3.

Table 2C.3 Standard enthalpies of formation and combustion of organic compounds at 298 K*

	$\Delta_f H^{\ominus}/(\text{kJ mol}^{-1})$	$\Delta_c H^{\ominus}/(\text{kJ mol}^{-1})$
Benzene, $C_6H_6(l)$	+49.0	−3268
Ethane, $C_2H_6(g)$	−84.7	−1560
Glucose, $C_6H_{12}O_6(s)$	−1274	−2808
Methane, $CH_4(g)$	−74.8	−890
Methanol, $CH_3OH(l)$	−238.7	−721

Per v_x mole X

Hess's law / Hess 定律

- The standard reaction enthalpy is the sum of the values of the individual reactions into which the overall reaction may be divided.

Example 2C.1

- The standard reaction enthalpy for the hydrogenation of propene,

$$CH_2{=}CHCH_3(g) + H_2(g) \rightarrow CH_3CH_2CH_3(g)$$

is $-124\,kJ\,mol^{-1}$. The standard reaction enthalpy for the combustion of propane,

$$CH_3CH_2CH_3(g) + 5\,O_2(g) \rightarrow 3\,CO_2(g) + 4\,H_2O(l)$$

is $-2220\,kJ\,mol^{-1}$. The standard reaction enthalpy for the formation of water,

$$H_2(g) + \tfrac{1}{2}\,O_2(g) \rightarrow H_2O(l)$$

is $-286\,kJ\,mol^{-1}$. Calculate the standard enthalpy of combustion of propene.

The solution The combustion reaction is

$$C_3H_6(g) + \tfrac{9}{2}\,O_2(g) \rightarrow 3\,CO_2(g) + 3\,H_2O(l)$$

This reaction can be recreated from the following sum:

	$\Delta_r H^{\ominus}/(kJ\,mol^{-1})$
$C_3H_6(g) + H_2(g) \rightarrow C_3H_8(g)$	−124
$C_3H_8(g) + 5\,O_2(g) \rightarrow 3\,CO_2(g) + 4\,H_2O(l)$	−2220
$H_2O(l) \rightarrow H_2(g) + \tfrac{1}{2}\,O_2(g)$	+286
$C_3H_6(g) + \tfrac{9}{2}\,O_2(g) \rightarrow 3\,CO_2(g) + 3\,H_2O(l)$	−2058

· Hess's law is the First Law applied to reaction enthalpies.

Hess's law / Hess 定律

- Calculate the standard enthalpy of hydrogenation of liquid benzene苯 from its standard enthalpy of combustion (−3268 kJ mol^{-1}) and the standard enthalpy of combustion of liquid cyclohexane环己烷 (−3920 kJ mol^{-1}). The standard enthalpy of combustion of hydrogen is -286 kJ mol^{-1}
-

1. Benzene combustion:

$$C_6H_6(l) + \tfrac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l) \quad \Delta_c H_1^{\ominus} = -3268\ \text{kJ} \cdot \text{mol}^{-1}$$

2. Cyclohexane combustion:

$$C_6H_{12}(l) + 9O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l) \quad \Delta_c H_2^{\ominus} = -3920\ \text{kJ} \cdot \text{mol}^{-1}$$

3. Hydrogen combustion:

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l) \quad \Delta_c H^{\ominus}(H_2, g) = -286\ \text{kJ} \cdot \text{mol}^{-1}$$

We need: $\text{Reaction}(1) + 3 \times \text{Reaction}(3) - \text{Reaction}(2)$ gives the hydrogenation reaction.

The hydrogenation reaction: $C_6H_6(l) + 3H_2(g) \rightarrow C_6H_{12}(l)$

$$\begin{aligned}\Delta_r H^{\ominus} &= \left[\Delta_c H^{\ominus}(C_6H_6, l) + 3\Delta_c H^{\ominus}(H_2, g)\right] - \Delta_c H^{\ominus}(C_6H_{12}, l) \\ &= \left[-3268 + 3(-286)\right] - (-3920) \\ &= \mathbf{-206}\ \text{kJ mol}^{-1}\end{aligned}$$

Standard enthalpy of formation

- Standard enthalpy of formation, $\Delta_f H^{\ominus}$ 标准生成焓, the standard reaction enthalpy for the formation of the compound from its elements单质 in their reference states. (标准摩尔生成焓$\Delta_f H_m^{\ominus}$)
- The **reference state** of an element is its most stable state at the specified temperature and 1 bar.
- At 298K, N->N_2(g), Hg->Hg(l), C->graphite

One exception: P->white phosphorus is not the most stable form but the most reproducible

- The standard enthalpies of formation of elements in their reference states are 0 at all temperatures

$$6\,C(s,graphite) + 3\,H_2(g) \rightarrow C_6H_6(l)$$

The standard enthalpy of formation of liquid benzene at 298 K is +49.0 kJ mol^{-1}

Table 2C.3 Standard enthalpies of formation and combustion of organic compounds at 298 K*

	$\Delta_f H^{\ominus}$/(kJ mol^{-1})	$\Delta_c H^{\ominus}$/(kJ mol^{-1})
Benzene, C_6H_6(l)	+49.0	−3268
Ethane, C_2H_6(g)	−84.7	−1560
Glucose, $C_6H_{12}O_6$(s)	−1274	−2808
Methane, CH_4(g)	−74.8	−890
Methanol, CH_3OH(l)	−238.7	−721

· The superscript $^{\ominus}$ denotes the standard state.

Standard enthalpy of formation

- The standard enthalpy of formation of ions in solution poses a special problem because it is not possible to prepare a solution of either cations or anions alone. This problem is overcome by defining one ion, conventionally the hydrogen, to have zero standard enthalpy of formation at all temperatures:

$$\Delta_f H^{\ominus}(H^+,aq) = 0 \qquad \text{Ions in solution [convention]} \qquad (2C.4)$$

Brief illustration 2C.2

If the enthalpy of formation of HBr(aq) is found to be $-122\,kJ\,mol^{-1}$, then the whole of that value is ascribed to the formation of $Br^-(aq)$, and $\Delta_f H^{\ominus}(Br^-,aq) = -122\,kJ\,mol^{-1}$. That value may then be combined with, for instance, the enthalpy of formation of AgBr(aq) to determine the value of $\Delta_f H^{\ominus}(Ag^+,aq)$, and so on. In essence, this definition adjusts the actual values of the enthalpies of formation of ions by a fixed value, which is chosen so that the standard value for one of them, $H^+(aq)$, is zero.

Standard enthalpy of formation

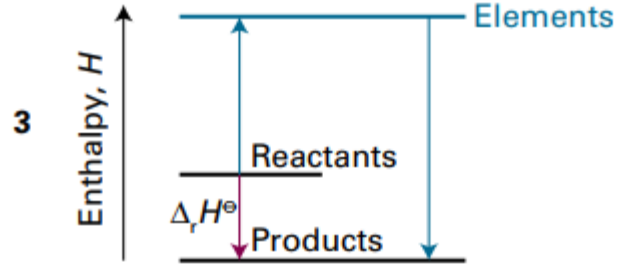
- The value of $\Delta_r H^{\ominus}$ for the overall reaction is the sum of these ‘unforming’ and forming enthalpies.
- ‘Unforming’ is the reverse of forming, the enthalpy of an unforming step is the negative of the enthalpy of formation
- stoichiometric numbers 化学计量数ν_J (as distinct from the stoichiometric coefficients化学计量系数) which are positive for products and negative for reactants.

$$\Delta_r H^{\circ} = \sum_{\text{Products}} \nu \Delta_f H^{\circ} - \sum_{\text{Reactants}} \nu \Delta_f H^{\ominus}$$

Standard reaction enthalpy [practical implementation]

(2C.5a)

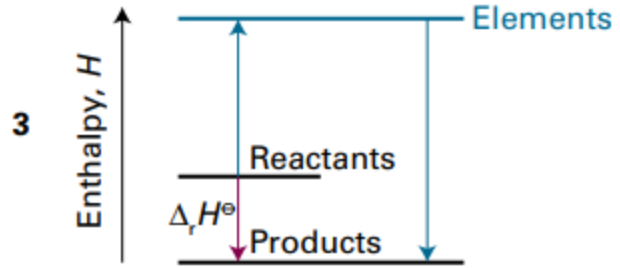
$$\Delta_r H^{\ominus} = \sum_J \nu_J \Delta_f H^{\ominus}(J)$$



Standard enthalpy of formation

$$\Delta_r H^\circ = \sum_{\text{Products}} \nu \Delta_f H^\circ - \sum_{\text{Reactants}} \nu \Delta_f H^\ominus$$

Standard reaction enthalpy [practical implementation] (2C.5a)



Brief illustration 2C.3

According to eqn 2C.5a, the standard enthalpy of the reaction $2\,HN_3(l) + 2\,NO(g) \rightarrow H_2O_2(l) + 4\,N_2(g)$ is calculated as follows:

$$\begin{aligned}\Delta_r H^\ominus &= \{\Delta_f H^\ominus(H_2O_2,l) + 4\Delta_f H^\ominus(N_2,g)\} \\ &\quad - \{2\Delta_f H^\ominus(HN_3,l) + 2\Delta_f H^\ominus(NO,g)\} \\ &= \{-187.78 + 4(0)\}\,\text{kJ mol}^{-1} \\ &\quad - \{2(264.0) + 2(90.25)\}\,\text{kJ mol}^{-1} \\ &= -896.3\,\text{kJ mol}^{-1}\end{aligned}$$

To use eqn 2C.5b, identify $\nu(HN_3) = -2$, $\nu(NO) = -2$, $\nu(H_2O_2) = +1$, and $\nu(N_2) = +4$, and then write

$$\begin{aligned}\Delta_r H^\ominus &= \Delta_f H^\ominus(H_2O_2,l) + 4\Delta_f H^\ominus(N_2,g) - 2\Delta_f H^\ominus(HN_3,l) \\ &\quad - 2\Delta_f H^\ominus(NO,g)\end{aligned}$$

which gives the same result.

The temperature dependence of reaction enthalpies

- The standard reaction enthalpies at different temperatures can be calculated from heat capacities and the reaction enthalpy at some other temperature.
- $dH = C_p dT$.

$$H(T_2) = H(T_1) + \int_{T_1}^{T_2} C_p \, dT \qquad (2C.6)$$

$$\Delta_r H^{\ominus}(T_2) = \Delta_r H^{\ominus}(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^{\ominus} \, dT \qquad \text{Kirchhoff's law} \qquad (2C.7a)$$

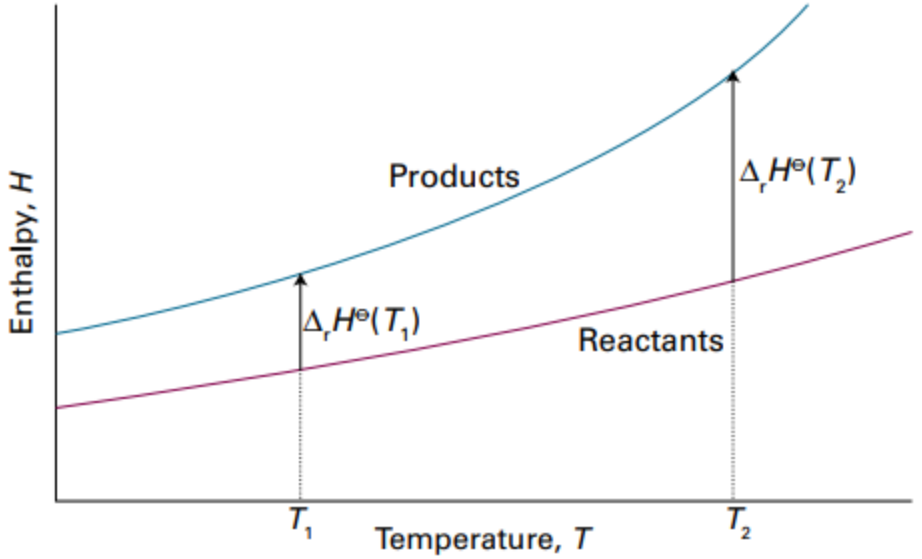


Figure 2C.1 When the temperature is increased, the enthalpy of the products and the reactants both increase, but may do so to different extents. In each case, the change in enthalpy depends on the heat capacities of the substances. The change in reaction enthalpy reflects the difference in the changes of the enthalpies of the products and reactants.

· DSC reports transition temperatures and enthalpies.

The temperature dependence of reaction enthalpies

$$H(T_2) = H(T_1) + \int_{T_1}^{T_2} C_p \, dT \qquad (2C.6)$$

$$\Delta_r H^{\ominus}(T_2) = \Delta_r H^{\ominus}(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^{\ominus} \, dT \qquad \text{Kirchhoff's law} \qquad (2C.7a)$$

- $\Delta_r C_p^{\ominus}$ is the difference of the molar heat capacities of products and reactants under standard conditions weighted by the stoichiometric coefficients that appear in the chemical equation

$$\Delta_r C_p^{\ominus} = \sum_{\text{Products}} \nu C_{p,m}^{\ominus} - \sum_{\text{Reactants}} \nu C_{p,m}^{\ominus} \qquad (2C.7b)$$

$$\Delta_r C_p^{\ominus} = \sum_{J} \nu_J C_{p,m}^{\ominus}(J) \qquad (2C.7c)$$

- If $\Delta_r C_p^{\ominus}$ is largely independent of temperature in the range T1 to T2,

$$\Delta_r H^{\ominus}(T_2) = \Delta_r H^{\ominus}(T_1) + \Delta_r C_p^{\ominus}(T_2 - T_1) \qquad \text{Integrated form of Kirchhoff's law} \qquad (2C.7d)$$

Using Kirchhoff's law

$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + \int_{T_1}^{T_2} \Delta_r C_p^\ominus \, dT \qquad \text{Kirchhoff's law} \qquad (2C.7a)$$

$$\Delta_r C_p^\ominus = \sum_{\text{Products}} \nu C_{p,m}^\ominus - \sum_{\text{Reactants}} \nu C_{p,m}^\ominus \qquad (2C.7b)$$

$$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + \Delta_r C_p^\ominus (T_2 - T_1) \qquad \text{Integrated form of Kirchhoff's law} \qquad (2C.7d)$$

Example 2C.2 Using Kirchhoff's law

The standard enthalpy of formation of $H_2O(g)$ at 298 K is $-241.82\,kJ\,mol^{-1}$. Estimate its value at 100 °C given the following values of the molar heat capacities at constant pressure: $H_2O(g)$: $33.58\,J\,K^{-1}\,mol^{-1}$; $H_2(g)$: $28.84\,J\,K^{-1}\,mol^{-1}$; $O_2(g)$: $29.37\,J\,K^{-1}\,mol^{-1}$. Assume that the heat capacities are independent of temperature.

Collect your thoughts When $\Delta_r C_p^\ominus$ is independent of temperature in the range T_1 to T_2, you can use the integrated form of the Kirchhoff equation, eqn 2C.7d. To proceed, write the chemical equation, identify the stoichiometric coefficients, and calculate $\Delta_r C_p^\ominus$ from the data.

The solution The reaction is $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$, so

$$\begin{aligned}\Delta_r C_p^\ominus &= C_{p,m}^\ominus(H_2O,g) - \{C_{p,m}^\ominus(H_2,g) + \tfrac{1}{2} C_{p,m}^\ominus(O_2,g)\} \\ &= -9.94\,J\,K^{-1}\,mol^{-1}\end{aligned}$$

It then follows that

$$\begin{aligned}\Delta_r H^\ominus(373\,K) &= -241.82\,kJ\,mol^{-1} + (75\,K) \\ &\quad \times (-9.94\,J\,K^{-1}\,mol^{-1}) = -242.6\,kJ\,mol^{-1}\end{aligned}$$

The enthalpy changes only slightly because ΔCp is small.

Using Kirchhoff's law

$$\Delta_r C_p^{\ominus} = \sum_{\text{Products}} \nu C_{p,\text{m}}^{\ominus} - \sum_{\text{Reactants}} \nu C_{p,\text{m}}^{\ominus} \qquad \text{(2C.7b)}$$

$$\Delta_r H^{\circ}(T_2) = \Delta_r H^{\circ}(T_1) + \Delta_r C_p^{\circ}(T_2 - T_1) \qquad \text{(2C.7d)}$$

Integrated form of Kirchhoff's law

- Estimate the standard enthalpy of formation of cyclohexane, $C_6H_{12}(l)$, at 400 K from the data of 298K (-156kJ·mol^{-1}) and heat capacity data given in the Resource section
-

- Molar constant-pressure heat capacities $C_{p,m}/(\mathrm{J \cdot mol^{-1} \cdot K^{-1}})$:
  - $\mathrm{C_6H_{12}(l)} : C_{p,m} = 156$
  - $\mathrm{C(s, graphite)} : C_{p,m} = 8.53$
  - $\mathrm{H_2(g)} : C_{p,m} = 28.82$

Reaction: $6\,\mathrm{C(graphite, s)} + 6\,\mathrm{H_2(g)} \rightarrow \mathrm{C_6H_{12}(l)}$

$$\begin{aligned}
\Delta_r C_p^{\ominus} &= C_{p,m}(\mathrm{C_6H_{12}, l}) - \left[6C_{p,m}(\mathrm{C, gr}) + 6C_{p,m}(\mathrm{H_2, g})\right] = -68.10\ \mathrm{J\,mol^{-1}\,K^{-1}} \\
\Delta_f H^{\ominus}(400\,\mathrm{K}) &= \Delta_f H^{\ominus}(298\,\mathrm{K}) + \Delta_r C_p^{\ominus}\,(400 - 298) \\
&= -156 + (-68.10 \times 10^{-3}) \times 102 \\
&= \mathbf{-163}\ \mathrm{kJ\,mol^{-1}}
\end{aligned}$$

Differential scanning calorimetry (DSC)

- A DSC measures the energy transferred as heat at constant pressure during a change.
- 'Differential': the sample is compared with an inert reference.

差示

- 'Scanning': the temperature is increased at a constant rate.

扫描

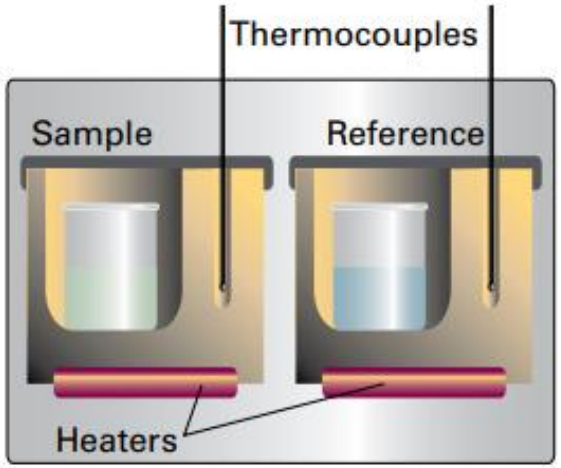


Figure 2C.2 A differential scanning calorimeter. The sample and a reference material are heated in separate but identical metal heat sinks. The output is the difference in power needed to maintain the heat sinks at equal temperatures as the temperature rises.

· DSC reports transition temperatures and enthalpies.

TOPIC 2C · FIG 2C.3

Reading a DSC thermogram

- The thermogram records heat flow against temperature.
- A peak (or step) marks a physical or chemical transition.
  峰
- The area under the peak gives the transition enthalpy.
  面积 = 焓变

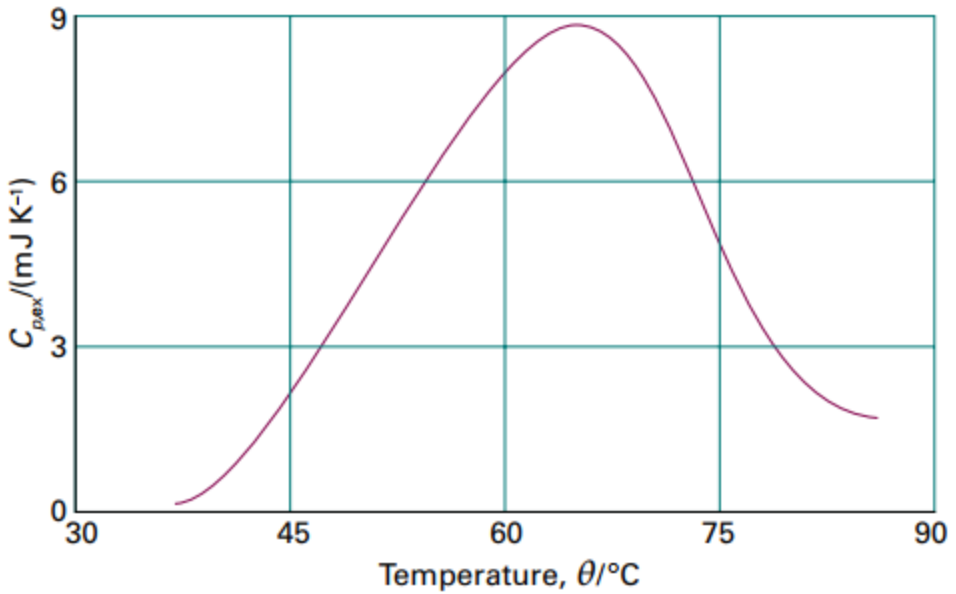


Figure 2C.3 A thermogram for the protein ubiquitin at pH = 2.45. The protein retains its native structure up to about 45 °C and then undergoes an endothermic conformational change. (Adapted from B. Chowdhry and S. LeHarne, *J. Chem. Educ.* 74, 236 (1997).)

· Calorimetry measures heat we cannot get from a table.

Isothermal titration calorimetry (ITC)

- A differential calorimetric method.
- Two cells: reference cell and sample cell, each fitted with heater.
- Titrant is injected step-wise into sample cell. Reaction releases or absorbs heat.
- Instrument adjusts heating power to keep sample cell at the same temperature as reference cell.
- Exothermic reaction → less power required; endothermic → more power required.
- Heat for each injection $q_i = P_i \Delta t$. Sum heat data and concentration change to obtain reaction enthalpy $\Delta_r H^{\ominus}$, binding constant and stoichiometry.

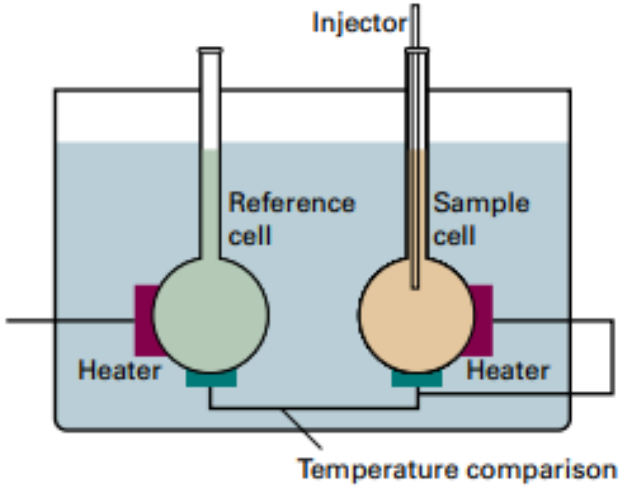


Figure 2C.4 A schematic diagram of the apparatus used for isothermal titration calorimetry.

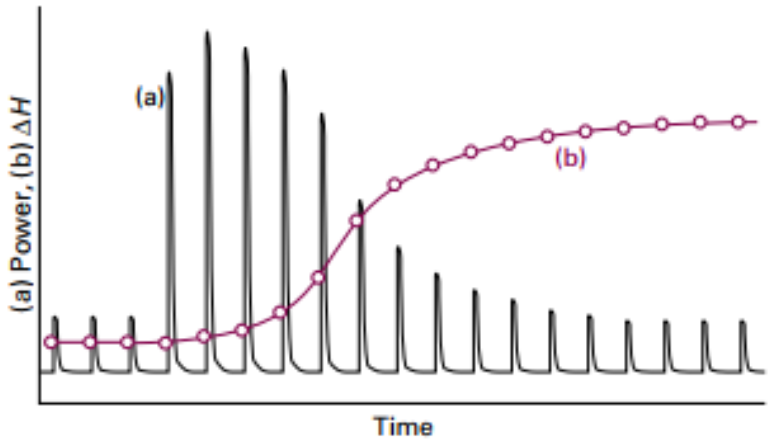


Figure 2C.5 (a) The record of the power applied as each injection is made, and (b) the sum of successive enthalpy changes in the course of the titration.

Check your understanding

If $\Delta_r C_p^\circ = 0$, then $\Delta_r H^\circ$:

- A. increases with T
- B. decreases with T
- C. is independent of T
- D. becomes zero

Answer · 答案: C — a zero ΔrCp° means ΔrH° does not change with temperature.

This is why small ΔCp values make enthalpy nearly constant.

Checklist of concepts

- ☐ 1. The **standard enthalpy of transition** is equal to the energy transferred as heat at constant pressure in the transition under standard conditions.
- ☐ 2. The **standard state** of a substance at a specified temperature is its pure form at 1 bar.
- ☐ 3. A **thermochemical equation** is a chemical equation and its associated change in enthalpy.
- ☐ 4. **Hess's law** states that the standard reaction enthalpy is the sum of the values for the individual reactions into which the overall reaction may be divided.
- ☐ 5. **Standard enthalpies of formation** are defined in terms of the reference states of elements.
- ☐ 6. The **reference state** of an element is its most stable state at the specified temperature and 1 bar.
- ☐ 7. The **standard reaction enthalpy** is expressed as the difference of the standard enthalpies of formation of products and reactants.
- ☐ 8. The temperature dependence of a reaction enthalpy is expressed by **Kirchhoff's law**.

Checklist of equations

Property	Equation	Comment	Equation number
The standard reaction enthalpy	$\Delta_r H^\ominus = \sum_{\text{Products}} \nu \Delta_f H^\ominus - \sum_{\text{Reactants}} \nu \Delta_f H^\ominus$	ν: stoichiometric coefficients; ν_J: (signed) stoichiometric numbers	2C.5
	$\Delta_r H^\ominus = \sum_J \nu_J \Delta_f H^\ominus(J)$		
Kirchhoff's law	$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + \int_{T_1}^{T_2} \Delta_r C_P^\ominus \, dT$		2C.7a
	$\Delta_r C_P^\ominus = \sum_J \nu_J C_{P,m}^\ominus(J)$		2C.7c
	$\Delta_r H^\ominus(T_2) = \Delta_r H^\ominus(T_1) + (T_2 - T_1)\Delta_r C_P^\ominus$	If $\Delta_r C_P^\ominus$ independent of temperature	2C.7d

TOPIC 2D

State Functions & Exact Differentials

状态函数与全微分

1. Path-independence · 路径无关
2. Exact and inexact differentials · 全微分与非全微分
3. How U depends on V and T · U 对 V、T 的依赖
4. The Joule–Thomson effect · 焦耳–汤姆逊效应

➤ Why do you need to know this material?

Thermodynamics has the power to provide relations between a variety of properties. This Topic introduces its key procedure, the manipulation of equations involving state functions.

➤ What is the key idea?

The fact that internal energy and enthalpy are state functions leads to relations between thermodynamic properties.

TOPIC 2D · KEY IDEA

State functions: path-independent

- A state function depends only on the current state, not on how it was reached.
  状态函数
- U, H, p, V and T are state functions.
- Work and heat are NOT state functions — they depend on the path.
  功、热不是

· State functions let you reason about a change without knowing the details.

The same ΔU by different paths

- Two paths connect the same initial and final states on the U(V,T) surface.
- Both start and end at the same U — ΔU is the same.
- But the work and heat taken along each path differ.

功、热不同

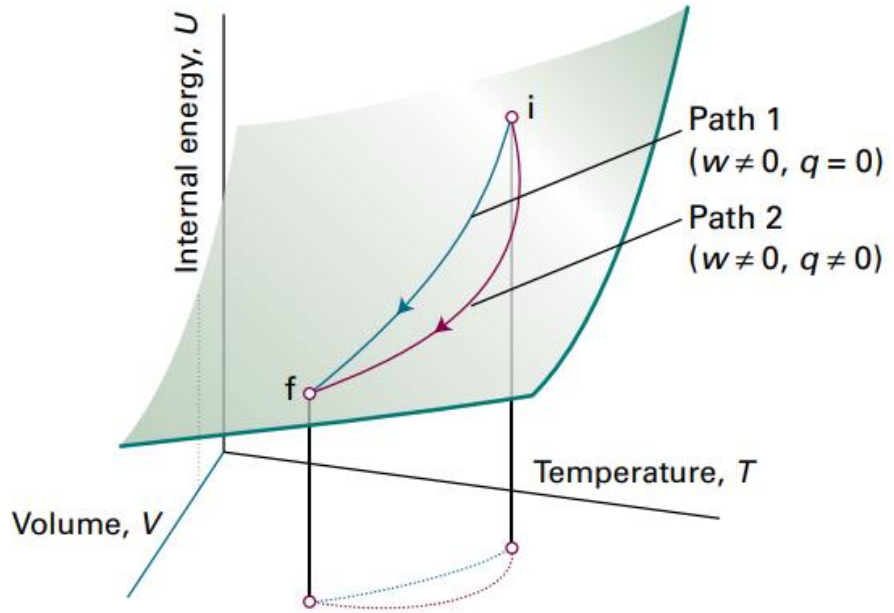


Figure 2D.1 As the volume and temperature of a system are changed, the internal energy changes. An adiabatic and a non-adiabatic path are shown as Path 1 and Path 2, respectively: they correspond to different values of *q* and *w* but to the same value of ΔU.

· The total change in U is a state-function change; the pieces along the way are not.

Exact and inexact differentials

- dU is an 'exact differential'全微分: an infinitesimal quantity that, when integrated, gives a result that is independent of the path between the initial and final states.

$$\Delta U = \int_{\mathrm{i}}^{\mathrm{f}} \mathrm{d}U$$

- When a system is heated, the total energy transferred as heat is the sum of all individual contributions at each point of the path

$$q = \int_{\mathrm{i,path}}^{\mathrm{f}} \mathrm{d}q$$

- The result of integration is q and not Δq. An adiabatic path has q = 0, whereas a non-adiabatic path between the same two states would have q ≠ 0
- An inexact differential非全微分 is an infinitesimal quantity that, when integrated, gives a result that depends on the path between the initial and final states. đq
- dw is an inexact differential. It is often written đw

· That is why we write dq, not Δq.

Calculating work, heat, and change in internal energy

Example 2D.1 Calculating work, heat, and change in internal energy

Consider a perfect gas inside a cylinder fitted with a piston. Let the initial state be T,V_i and the final state be T,V_f. The change of state can be brought about in many ways, of which the two simplest are the following:

- Path 1, in which there is free expansion against zero external pressure;
- Path 2, in which there is reversible, isothermal expansion.

Calculate w, q, and ΔU for each process.

Collect your thoughts To find a starting point for a calculation in thermodynamics, it is often a good idea to go back to first principles and to look for a way of expressing the quantity to be calculated in terms of other quantities that are easier to calculate. It is argued in Topic 2B that the internal energy of a perfect gas depends only on the temperature and is independent of the volume those molecules occupy, so for any isothermal change, $\Delta U = 0$. Also, $\Delta U = q + w$ in general. To solve the problem you need to combine the two expressions, selecting the appropriate expression for the work done from the discussion in Topic 2A.

Calculating work, heat, and change in internal energy

Example 2D.1 Calculating work, heat, and change in internal energy

Consider a perfect gas inside a cylinder fitted with a piston. Let the initial state be T,V_i and the final state be T,V_f. The change of state can be brought about in many ways, of which the two simplest are the following:

- Path 1, in which there is free expansion against zero external pressure;
- Path 2, in which there is reversible, isothermal expansion.

Calculate w, q, and ΔU for each process.

The solution Because $\Delta U = 0$ for both paths and $\Delta U = q + w$, in each case $q = -w$. The work of free expansion is zero (eqn 2A.7 of Topic 2A, $w = 0$); so in Path 1, $w = 0$ and therefore $q = 0$ too. For Path 2, the work is given by eqn 2A.9 of Topic 2A ($w = -nRT \ln(V_f/V_i)$) and consequently $q = nRT \ln(V_f/V_i)$.

Calculating work, heat, and change in internal energy

- Calculate the values of q, w, and ΔU for an irreversible isothermal expansion of a perfect gas against a constant non-zero external pressure
-

For a perfect-gas isothermal process: $\boldsymbol{\Delta U = 0}$.

Irreversible work against constant external pressure:

$$w = -p_{\text{ex}}(V_f - V_i)$$

From first law $\Delta U = q + w = 0$:

$$q = -w = p_{\text{ex}}(V_f - V_i)$$

3. 热力学第一定律 $\Delta U = q + w$，$\Delta U = 0$，所以：

$$q = -w$$

气体对外做功，系统从环境吸热。

> 对比记忆：
>
> - 可逆等温膨胀：$w = -nRT\ln(V_f/V_i)$，$q = -w$，$\Delta U = 0$
> - 恒外压不可逆等温膨胀：$w = -p_{\text{ex}}\Delta V$，$q = -w$，$\Delta U = 0$

Changes in internal energy

- U can be regarded as a function of V, T, and p. Choosing the values of two of the variables fixes the value of the third, eg. pV=nRT

$$\mathrm{d}U = \left(\frac{\partial U}{\partial V}\right)_T \mathrm{d}V + \left(\frac{\partial U}{\partial T}\right)_V \mathrm{d}T \qquad \text{(2D.3)}$$

General expression for a change in U with T and V

- in a closed system of constant composition, any infinitesimal change in the internal
energy is proportional to the infinitesimal changes of volume and temperature

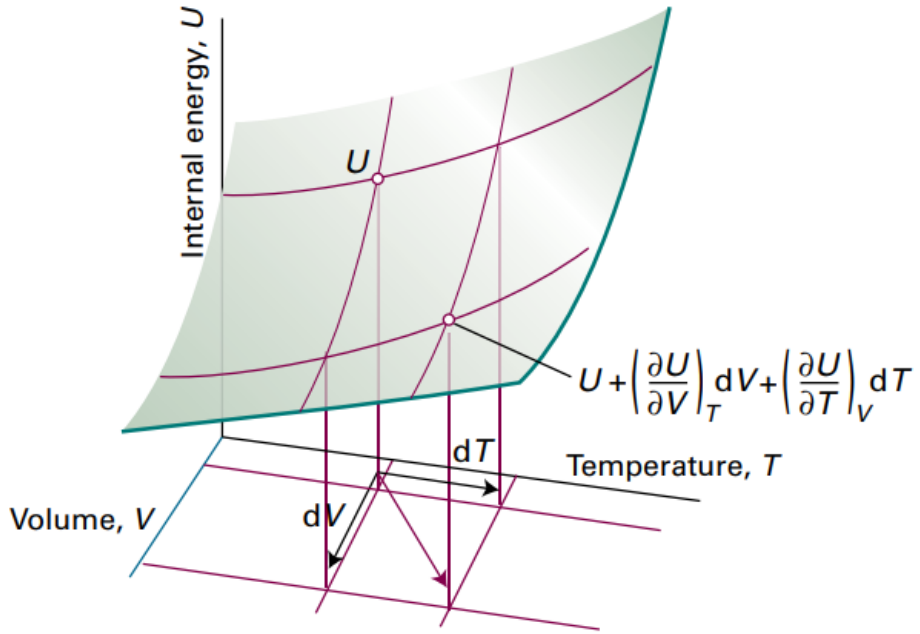


Figure 2D.2 An overall change in U, which is denoted dU, arises when both V and T are allowed to change. If second-order infinitesimals are ignored, the overall change is the sum of changes for each variable separately.

· That is why we write dq, not Δq.

The internal pressure π_T

- $\pi_T = (\partial U/\partial V)_T$ is the slope of U against V at constant T.
  内压
- It measures how internal energy responds to a volume change.
- $C_V = (\partial U/\partial V)_V$ $dU = \pi_T dV + C_V dT$

- For a perfect gas, $\pi = 0$: U is independent of volume.

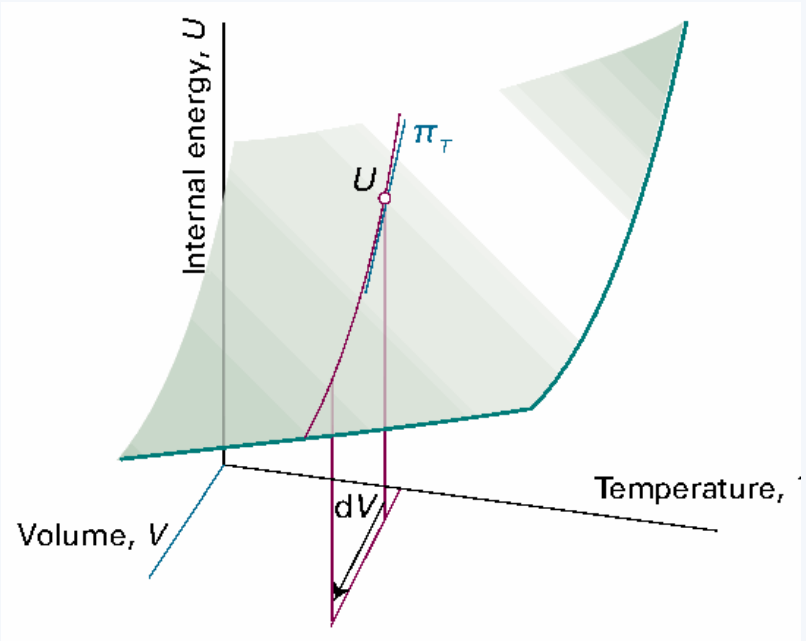


Fig 2D.3 The internal pressure

· The internal pressure is the 'pressure' that energy exerts on volume.

U of a perfect gas vs volume

- For a perfect gas, U is the same at every volume (at a given T).

U 与 V 无关

- Attractions make U fall as V increases; repulsions raise it.

引力/斥力

- The perfect gas line is flat: π = 0.

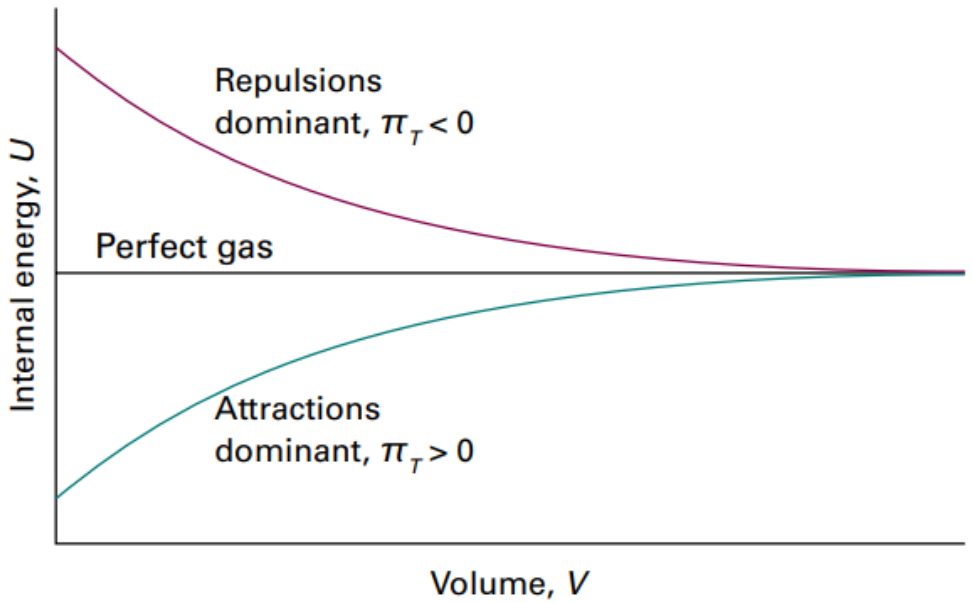


Figure 2D.4 For a perfect gas, the internal energy is independent of the volume (at constant temperature). If attractions are dominant in a real gas, the internal energy increases with volume because the molecules become farther apart on average. If repulsions are dominant, the internal energy decreases as the gas expands.

· This justifies using ΔU = Cv ΔT for gases.

The internal pressure π_T

- when the pressure rather than the volume of the system is kept constant.
- $dU = \pi_T dV + C_V dT$

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

- the right in this expression is the slope of the plot of volume against temperature (at constant pressure). This property is normally tabulated as the expansion coefficient膨胀系数, α, of a substance, which is defined as

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$$

Expansion coefficient [definition] (2D.6)

A large value of α ~ the volume of the sample responds strongly to changes in temperature.

The internal pressure π_T

- when the pressure rather than the volume of the system is kept constant.
- $dU = \pi_T dV + C_V dT$

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

- isothermal compressibility等温压缩系数, κ_T (kappa)as

$$\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$$

Isothermal compressibility [definition] (2D.7)

- a measure of the fractional change in volume when the pressure is increased

the negative sign ensures that the compressibility is a positive quantity, because an increase of pressure, implying a positive dp, brings about a reduction of volume, a negative dV

· The internal pressure is the 'pressure' that energy exerts on volume.

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$$

Expansion coefficient [definition] (2D.6)

Expansion coefficient of a perfect gas

- Derive an expression for the expansion coefficient of a perfect gas
- *Collect your thoughts* The expansion coefficient is defined in eqn 2D.6. To use this expression, you need to substitute the expression for V in terms of T obtained from the equation of state for the gas. As implied by the subscript in eqn 2D.6, the pressure, p, is treated as a constant.

The solution Because $pV = nRT$, write

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p \overset{pV=nRT}{=} \frac{1}{V}\left(\frac{\partial (nRT/p)}{\partial T}\right)_p = \frac{1}{V} \times \frac{nR}{p} = \frac{nR}{pV} \overset{pV=nRT}{=} \frac{nR}{nRT} = \frac{1}{T}$$

The physical interpretation of this result is that the higher the temperature, the less responsive is the volume of a perfect gas to a change in temperature.

This connects gas properties to the First-Law state functions.

$$\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$$ Isothermal compressibility [definition] (2D.7)

Isothermal compressibility of a perfect gas

- Derive an expression for the isothermal compressibility of a perfect gas.
-

Definition:

$$\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$$

For perfect gas $V = \frac{nRT}{p}$:

$$\left(\frac{\partial V}{\partial p}\right)_T = -\frac{nRT}{p^2}$$

$$\kappa_T = -\frac{1}{V}\left(-\frac{nRT}{p^2}\right) = \frac{nRT}{Vp^2} = \frac{pV}{Vp^2} = \frac{\mathbf{1}}{\boldsymbol{p}}$$

- $\boldsymbol{\kappa_T = \frac{1}{p}}$: The higher the pressure, the smaller κ_T, so the gas is harder to compress; the gas is more compressible at low pressure.
- $\boldsymbol{\alpha = \frac{1}{T}}$: The higher the temperature, the smaller α, so the relative volume change induced by one-degree temperature rise is smaller at high temperature.

This connects gas properties to the First-Law state functions.

The internal pressure π_T

- when the pressure rather than the volume of the system is kept constant.
- $dU = \pi_T dV + C_V dT$

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V \qquad \alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$$

$$\boxed{\left(\frac{\partial U}{\partial T}\right)_p = \alpha \pi_T V + C_V}$$

- Perfect gas $\pi_T = 0$, $\left(\frac{\partial U}{\partial T}\right)_p = C_V$ (2D.9)

$$C_p - C_V = \overbrace{\left(\frac{\partial H}{\partial T}\right)_p}^{\text{Definition of } C_p} - \overbrace{\left(\frac{\partial U}{\partial T}\right)_p}^{\text{eqn 2D.9}}$$

$H = U + pV = U + nRT$

$$\boxed{C_p - C_V = \left(\frac{\partial (U + nRT)}{\partial T}\right)_p - \left(\frac{\partial U}{\partial T}\right)_p = nR}$$

Perfect gas

$$\boxed{C_p - C_V = \frac{\alpha^2 T V}{\kappa_T}}$$

Any substance

A large value of α ~ the volume of the sample responds strongly to changes in temperature.

C_p and C_v

$$C_p - C_V = \left(\frac{\partial(U+nRT)}{\partial T}\right)_p - \left(\frac{\partial U}{\partial T}\right)_p = nR$$

Perfect gas

$$C_p - C_V = \frac{\alpha^2 TV}{\kappa_T}$$

Any substance

Brief illustration 2D.1

The expansion coefficient and isothermal compressibility of water at 25 °C are given in Table 2D.1 as $2.1 \times 10^{-4}\,K^{-1}$ and $49.0 \times 10^{-6}\,bar^{-1}$ ($4.90 \times 10^{-10}\,Pa^{-1}$), respectively. The molar volume of water at that temperature, $V_m = M/\rho$ (where ρ is the mass density), is $18.1\,cm^3\,mol^{-1}$ ($1.81 \times 10^{-5}\,m^3\,mol^{-1}$). Therefore, from eqn 2D.11, the difference in molar heat capacities (which is given by using V_m in place of V) is

$$C_{p,m} - C_{V,m} = \frac{(2.1\times10^{-4}\,K^{-1})^2 \times (298\,K) \times (1.81\times10^{-5}\,m^3\,mol^{-1})}{4.90\times10^{-10}\,Pa^{-1}}$$

$$= 0.485\,Pa\,m^3\,K^{-1}\,mol^{-1} = 0.485\,J\,K^{-1}\,mol^{-1}$$

For water, $C_{p,m} = 75.3\,J\,K^{-1}\,mol^{-1}$, so $C_{V,m} = 74.8\,J\,K^{-1}\,mol^{-1}$. In some cases, the two heat capacities differ by as much as 30 per cent.

Table 2D.1 Expansion coefficients (α) and isothermal compressibilities (κ_T) at 298 K*

	$\alpha/(10^{-4}\,K^{-1})$	$\kappa_T/(10^{-6}\,bar^{-1})$
Liquids:		
Benzene	12.4	90.9
Water	2.1	49.0
Solids:		
Diamond	0.030	0.185
Lead	0.861	2.18

expansion coefficients α of liquids and solids are small, Cp ≈ Cv, κ_T might also be small?

Changes in enthalpy 焓变

- dH is an exact differential全微分, H is the function of p and T

Consider a closed system of constant composition. Because H is a function of p and T, when these two quantities change by an infinitesimal amount, the enthalpy changes by

$$dH = \left(\frac{\partial H}{\partial p}\right)_T dp + \left(\frac{\partial H}{\partial T}\right)_p dT$$

The second partial derivative is C_p. The task at hand is to express $(\partial H/\partial p)_T$ in terms of recognizable quantities. If the enthalpy is constant, then $dH = 0$ and

$$\left(\frac{\partial H}{\partial p}\right)_T dp = -C_p dT \quad \text{at constant } H$$

Division of both sides by dp then gives

$$\left(\frac{\partial H}{\partial p}\right)_T = -C_p\left(\frac{\partial T}{\partial p}\right)_H = -C_p\mu$$

焦耳 -汤姆孙系数

where the **Joule–Thomson coefficient**, μ (mu), is defined as

$$\mu = \left(\frac{\partial T}{\partial p}\right)_H$$

Joule–Thomson coefficient [definition] (2D.12)

It follows that

$$dH = -\mu C_p dp + C_p dT \quad (2D.13)$$

The variation of enthalpy with temperature and pressure

Changes in enthalpy 焓变

$$dH = -\mu C_p dp + C_p dT \quad (2D.13)$$

The variation of enthalpy with temperature and pressure

Brief illustration 2D.2

The Joule–Thomson coefficient for nitrogen at 298 K and 1 atm (Table 2D.2) is +0.27 K bar^{-1}. (Note that μ is an intensive property.) It follows that the change in temperature the gas undergoes when its pressure changes by −10 bar under isenthalpic conditions is

$$\Delta T \approx \mu \Delta p = +(0.27\ \text{K bar}^{-1}) \times (-10\ \text{bar}) = -2.7\,\text{K}$$

Table 2D.2 Inversion temperatures (T_i), normal freezing (T_f) and boiling (T_b) points, and Joule–Thomson coefficients (μ) at 1 atm and 298 K*

	T_i/K	T_f/K	T_b/K	μ/(K atm^{-1})
Ar	723	83.8	87.3	
CO_2	1500	194.7	+1.10	+1.11 at 300 K
He	40	4.2	4.22	−0.062
N_2	621	63.3	77.4	+0.27

A large value of α ~ the volume of the sample responds strongly to changes in temperature.

The Joule–Thomson effect

- $$\mu = \left(\frac{\partial T}{\partial p}\right)_H$$ Joule–Thomson coefficient [definition]
- The process is isenthalpic (constant enthalpy).
  等焓过程 $\Delta T \approx \mu \Delta p$
- A gas expands through a porous plug from high to low pressure.
- The change of temperature as a result of isenthalpic expansion is called the Joule–Thomson effect焦耳-汤姆孙效应

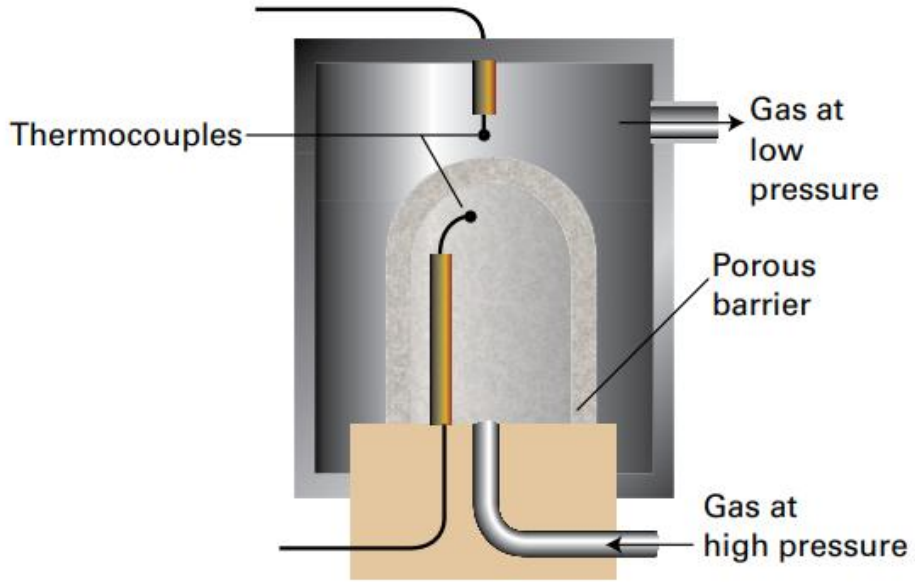


Figure 2D.6 The apparatus used for measuring the Joule–Thomson effect. The gas expands through the porous barrier, which acts as a throttle, and the whole apparatus is thermally insulated. As explained in the text, this arrangement corresponds to an isenthalpic expansion (expansion at constant enthalpy). Whether the expansion results in a heating or a cooling of the gas depends on the conditions.

· The Joule–Thomson effect underlies liquefaction液化 and refrigeration制冷.

How is that done? 2D.2 Establishing that the expansion is isenthalpic

Because all changes to the gas occur adiabatically, $q = 0$ and, consequently, $\Delta U = w$.

Step 1 *Calculate the total work*

Consider the work done as the gas passes through the barrier by focusing on the passage of a fixed amount of gas from the high pressure side, where the pressure is p_i, the temperature T_i, and the gas occupies a volume V_i (Fig. 2D.8). The gas emerges on the low pressure side, where the same amount of gas has a pressure p_f, a temperature T_f, and occupies a volume V_f. The gas on the left is compressed isothermally by the upstream gas acting as a piston. The relevant pressure is p_i and the volume changes from V_i to 0; therefore, the work done on the gas is

$$w_1 = -p_i(0 - V_i) = p_i V_i$$

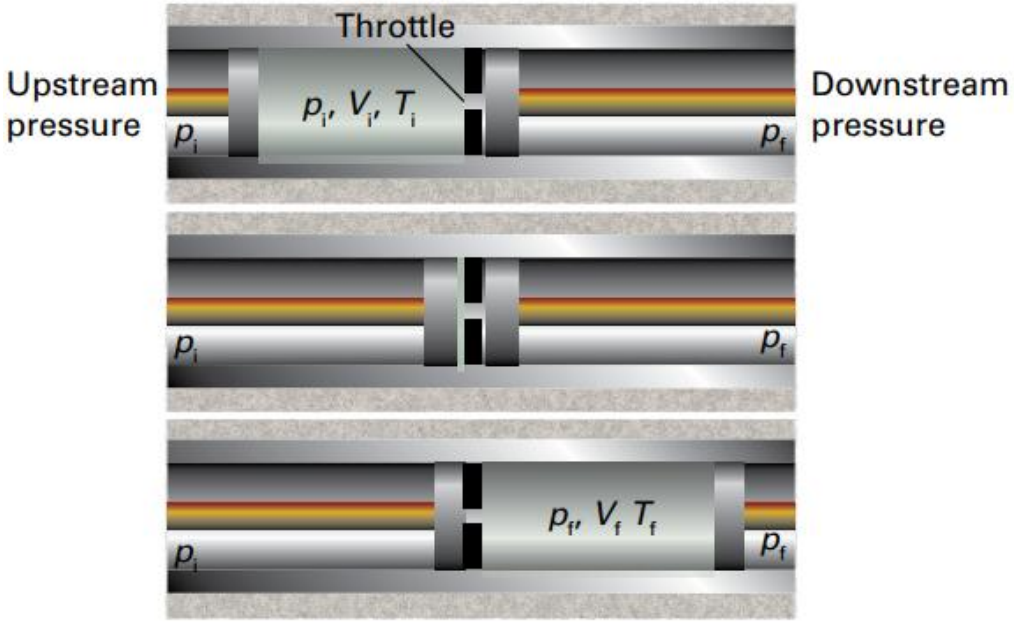


Figure 2D.8 The thermodynamic basis of Joule–Thomson expansion. The pistons represent the upstream and downstream gases, which maintain constant pressures either side of the throttle. The transition from the top diagram to the bottom diagram, which represents the passage of a given amount of gas through the throttle, occurs without change of enthalpy.

The gas expands isothermally on the right of the barrier (but possibly at a different constant temperature) against the pressure p_f provided by the downstream gas acting as a piston to be driven out. The volume changes from 0 to V_f, so the work done on the gas in this stage is

$$w_2 = -p_f(V_f - 0) = -p_f V_f$$

The total work done on the gas is the sum of these two quantities, or

$$w = w_1 + w_2 = p_i V_i - p_f V_f$$

Step 2 *Calculate the change in internal energy*

It follows that the change of internal energy of the gas as it moves adiabatically from one side of the barrier to the other is

$$U_f - U_i = w = p_i V_i - p_f V_f$$

Step 3 *Calculate the initial and final enthalpies*

Reorganization of the preceding expression, and noting that $H = U + pV$, gives

$$U_f + p_f V_f = U_i + p_i V_i \text{ or } H_f = H_i$$

Therefore, the expansion occurs without change of enthalpy.

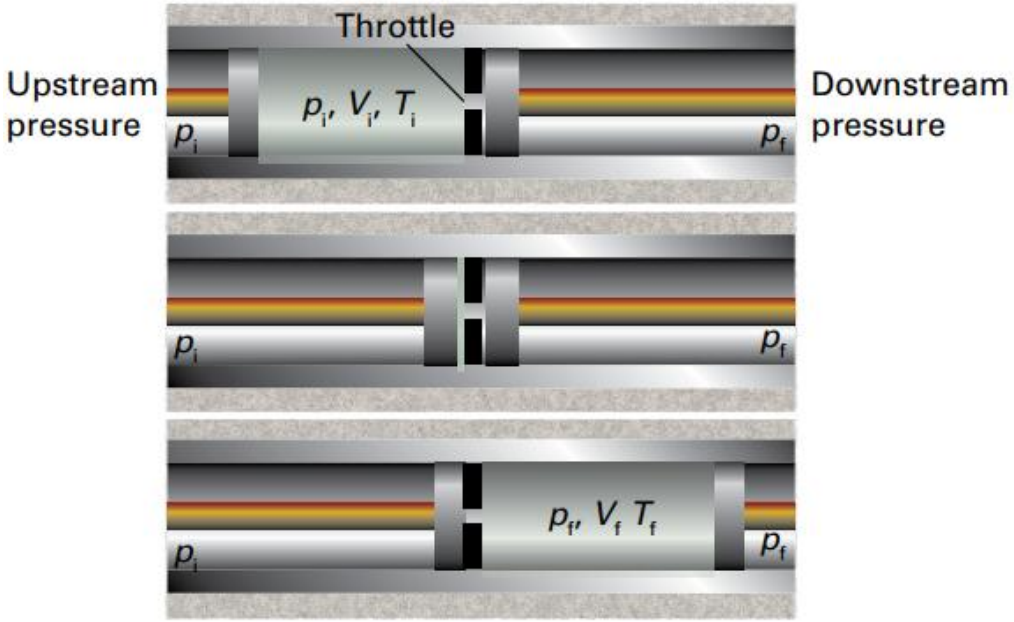


Figure 2D.8 The thermodynamic basis of Joule–Thomson expansion. The pistons represent the upstream and downstream gases, which maintain constant pressures either side of the throttle. The transition from the top diagram to the bottom diagram, which represents the passage of a given amount of gas through the throttle, occurs without change of enthalpy.

TOPIC 2D · SELF-TEST

Check your understanding

Which quantity is path-dependent?

- A. ΔU
- B. ΔH
- C. w
- D. All of the above

Answer · 答案: C — work (and heat) depend on the path; ΔU and ΔH are state functions.

State functions vs path functions is the crux of this topic.

Checklist of concepts

- ☐ 1. The quantity dU is an exact differential, dw and dq are not.
- ☐ 2. The change in internal energy may be expressed in terms of changes in temperature and volume.
- ☐ 3. The **internal pressure** is the variation of internal energy with volume at constant temperature.
- ☐ 4. **Joule's experiment** showed that the internal pressure of a perfect gas is zero.
- ☐ 5. The change in internal energy with pressure and temperature is expressed in terms of the internal pressure and the heat capacity and leads to a general expression for the relation between heat capacities.
- ☐ 6. The **Joule–Thomson effect** is the change in temperature of a gas when it undergoes isenthalpic expansion.

Checklist of equations

Property	Equation	Comment	Equation number
Change in $U(V,T)$	$dU = (\partial U/\partial V)_T dV + (\partial U/\partial T)_V dT$	Constant composition	2D.3
Internal pressure	$\pi_T = (\partial U/\partial V)_T$	Definition; for a perfect gas, $\pi_T = 0$	2D.4
Change in $U(V,T)$	$dU = \pi_T dV + C_V dT$	Constant composition	2D.5
Expansion coefficient	$\alpha = (1/V)(\partial V/\partial T)_p$	Definition	2D.6
Isothermal compressibility	$\kappa_T = -(1/V)(\partial V/\partial p)_T$	Definition	2D.7
Relation between heat capacities	$C_p - C_V = nR$	Perfect gas	2D.10
	$C_p - C_V = \alpha^2 TV/\kappa_T$		2D.11
Joule–Thomson coefficient	$\mu = (\partial T/\partial p)_H$	For a perfect gas, $\mu = 0$	2D.12
Change in $H(p,T)$	$dH = -\mu C_p dp + C_p dT$	Constant composition	2D.13

TOPIC 2E

Adiabatic Changes

绝热变化

➤ Why do you need to know this material?

Adiabatic processes complement isothermal processes, and are used in the discussion of the Second Law of thermodynamics.

➤ What is the key idea?

The temperature of a perfect gas falls when it does work in an adiabatic expansion.

1. Adiabatic means q = 0 · 绝热
2. Reversible adiabatic relations · 可逆绝热关系
3. Temperature and pressure changes · 温度与压力变化

Adiabatic changes: no heat

- Two steps, T_iV_i->T_iV_f->T_fV_f
- Provided the heat capacity is independent of temperature $\Delta U = (T_f - T_i)C_V = C_V\Delta T$
- An adiabatic process transfers no energy as heat (q = 0).
- So ΔU = w_{ad} — all the energy change is work.

Ad denotes adiabatic process

$$w_{ad} = C_V\Delta T$$

Work of adiabatic change [perfect gas]

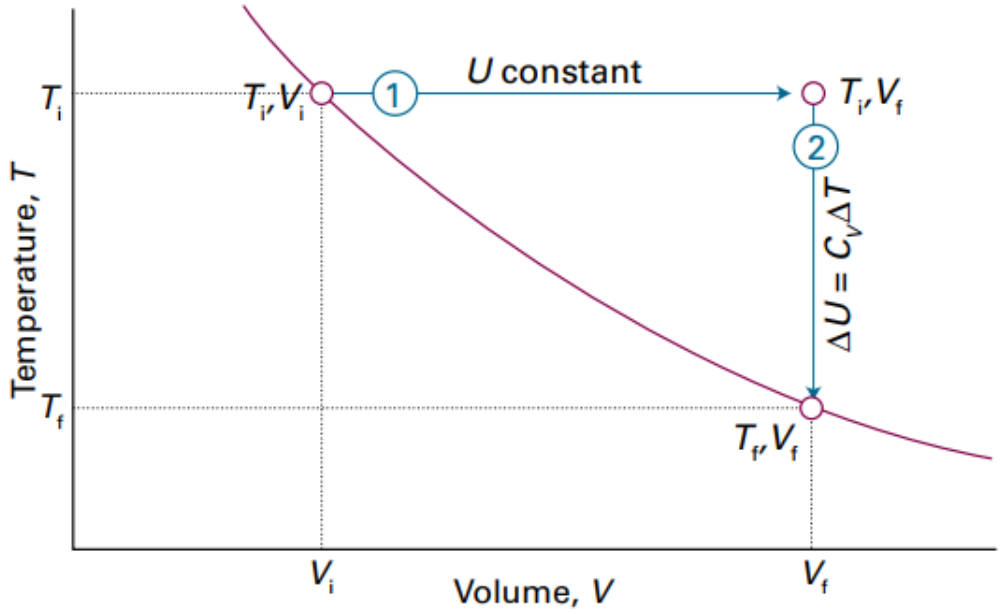


Figure 2E.1 To achieve a change of state from one temperature and volume to another temperature and volume, treat the overall change as composed of two steps. In the first step, the system expands at constant temperature; there is no change in internal energy if the system consists of a perfect gas. In the second step, the temperature of the system is reduced at constant volume. The overall change in internal energy is the sum of the changes for the two steps.

· An adiabatic wall blocks heat, but work can still be done.

Adiabatic changes

How is that done? 2E.1 Deriving an expression for the temperature change in a reversible adiabatic expansion

Consider a stage in a reversible adiabatic expansion of a perfect gas when the pressure inside and out is p. When considering reversible processes, it is usually appropriate to consider infinitesimal changes in the conditions, because pressures and temperatures typically change during the process. Then follow these steps.

Step 1 *Write an expression relating temperature and volume changes*

The work done when the gas expands reversibly by dV is $dw = -pdV$. This expression applies to any reversible change, including an adiabatic change, so specifically $dw_{ad} = -pdV$. Therefore, because $dq = 0$ for an adiabatic change, $dU = dw_{ad}$ (the infinitesimal version of $\Delta U = w_{ad}$).

For a perfect gas, $dU = C_V dT$ (the infinitesimal version of $\Delta U = C_V \Delta T$). Equating these expressions for dU gives

$$C_V dT = -pdV$$

Because the gas is perfect, p can be replaced by nRT/V to give $C_V dT = -(nRT/V)dV$ and therefore

$$\frac{C_V dT}{T} = -\frac{nRdV}{V}$$

Adiabatic changes

Step 2 *Integrate the expression to find the overall change*

To integrate this expression, ensure that the limits of integration match on each side of the equation. Note that T is equal to T_i when V is equal to V_i, and is equal to T_f when V is equal to V_f at the end of the expansion. Therefore,

$$C_V \int_{T_i}^{T_f} \frac{dT}{T} = -nR \int_{V_i}^{V_f} \frac{dV}{V}$$

where C_V is taken to be independent of temperature. Use Integral A.2 in each case, and obtain

$$C_V \ln \frac{T_f}{T_i} = -nR \ln \frac{V_f}{V_i}$$

Step 3 *Simplify the expression*

Because $\ln(x/y) = -\ln(y/x)$, the preceding expression rearranges to

$$\frac{C_V}{nR} \ln \frac{T_f}{T_i} = \ln \frac{V_i}{V_f}$$

Next, note that $C_V/nR = C_{V,m}/R = c$ and use $\ln x^a = a \ln x$ to obtain

$$\ln \left(\frac{T_f}{T_i} \right)^c = \ln \frac{V_i}{V_f}$$

This relation implies that $(T_f/T_i)^c = (V_i/V_f)$ and, upon rearrangement,

$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{1/c} \qquad c = C_{V,m}/R \qquad \text{(2E.2a)}$$

Temperature change [reversible adiabatic expansion, perfect gas]

By raising each side of this expression to the power c and reorganizing it slightly, an equivalent expression is

$$V_i T_i^c = V_f T_f^c \qquad c = C_{V,m}/R \qquad \text{(2E.2b)}$$

Temperature change [reversible adiabatic expansion, perfect gas]

This result is often summarized in the form $VT^c = \text{constant}$.

Adiabatic changes

$$T_f = T_i\left(\frac{V_i}{V_f}\right)^{1/c} \qquad c = C_{V,m}/R \tag{2E.2a}$$

Temperature change [reversible adiabatic expansion, perfect gas]

Brief illustration 2E.1

Consider the adiabatic, reversible expansion of 0.020 mol Ar, initially at 25 °C, from 0.50 dm^3 to 1.00 dm^3. The molar heat capacity of argon at constant volume is 12.47 $J K^{-1} mol^{-1}$, so $c = 1.501$. Therefore, from eqn 2E.2a,

$$T_f = (298\,K)\times\left(\frac{0.50\,dm^3}{1.00\,dm^3}\right)^{1/1.501} = 188\,K$$

It follows that $\Delta T = -110\,K$, and therefore, from eqn 2E.1, that

$$w_{ad} = \{(0.020\,mol)\times(12.47\,J\,K^{-1}\,mol^{-1})\}\times(-110\,K) = -27\,J$$

Note that temperature change is independent of the amount of gas but the work is not.

The change in pressure

$$T_f = T_i\left(\frac{V_i}{V_f}\right)^{1/c} \qquad c = C_{V,m}/R \tag{2E.2a}$$

Temperature change [reversible adiabatic expansion, perfect gas]

How is that done? 2E.2 Deriving the relation between pressure and volume for a reversible adiabatic expansion

The initial and final states of a perfect gas satisfy the perfect gas law regardless of how the change of state takes place, so $pV = nRT$ can be used to write

$$\frac{p_i V_i}{p_f V_f} = \frac{T_i}{T_f}$$

However, $T_i/T_f = (V_f/V_i)^{1/c}$ (eqn 2E.2a). Therefore,

$$\frac{p_i V_i}{p_f V_f} = \left(\frac{V_f}{V_i}\right)^{1/c}, \quad \text{so} \quad \frac{p_i}{p_f}\left(\frac{V_i}{V_f}\right)^{\frac{1}{c}+1} = 1$$

For a perfect gas $C_{p,m} - C_{V,m} = R$ (Topic 2B). It follows that

$$\frac{1}{c} + 1 = \frac{1+c}{c} = \frac{R + C_{V,m}}{C_{V,m}} = \frac{C_{p,m}}{C_{V,m}} = \gamma$$

and therefore that

$$\frac{p_i}{p_f}\left(\frac{V_i}{V_f}\right)^{\gamma} = 1$$

which rearranges to

$$p_f V_f^{\gamma} = p_i V_i^{\gamma} \tag{2E.3}$$

Pressure change [reversible adiabatic expansion, perfect gas]

This result is commonly summarized in the form $pV^{\gamma} =$ constant.

Adiabats versus isotherms

- The curves of pressure versus volume for adiabatic change are known as adiabats绝热线

Because γ > 1, an adiabat falls more steeply ($p \propto 1/V^{\gamma}$) than the corresponding isotherm ($p \propto 1/V$).

- In an isothermal expansion heat flows in and maintains T.
- In an adiabatic expansion no heat flows, so T falls and p drops more.

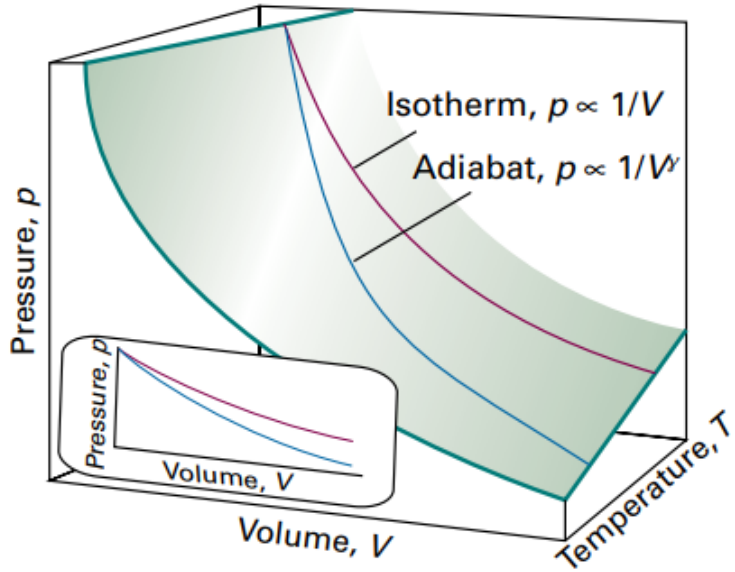


Figure 2E.2 An adiabat depicts the variation of pressure with volume when a gas expands adiabatically and, in this case, reversibly. Note that the pressure declines more steeply for an adiabat than it does for an isotherm because in an adiabatic change the temperature falls.

c = Cv,m/R; for a monatomic perfect gas γ = 5/3.

· Compare the two curves — the adiabat drops more steeply.

Adiabats versus isotherms

Brief illustration 2E.2

When a sample of argon (for which $\gamma = \frac{5}{3}$) at 100 kPa expands reversibly and adiabatically to twice its initial volume the final pressure will be

$$p_f = \left(\frac{V_i}{V_f}\right)^{\gamma} p_i = \left(\frac{1}{2}\right)^{5/3} \times (100\,\text{kPa}) = 31\,\text{kPa}$$

For an isothermal expansion in which the volume doubles the final pressure would be 50 kPa.

c = Cv,m/R; for a monatomic perfect gas γ = 5/3.

· Compare the two curves — the adiabat drops more steeply.

Check your understanding

In a reversible adiabatic expansion of a perfect gas, which is zero?

- A. Work
- B. ΔU
- C. Heat (q)
- D. ΔT

Answer · 答案: C — the process is adiabatic, so q = 0.

All the energy change appears as work, so the gas cools.

Checklist of concepts

☐ 1. The temperature of a gas falls when it undergoes an adiabatic expansion in which work is done.

☐ 2. An **adiabat** is a curve showing how pressure varies with volume in an adiabatic process.

Checklist of equations

Property	Equation	Comment	Equation number
Work of adiabatic expansion	$w_{ad} = C_V \Delta T$	Perfect gas	2E.1
Final temperature	$T_f = T_i (V_i / V_f)^{1/c}$	Perfect gas, reversible adiabatic expansion	2E.2a
	$c = C_{V,m} / R$		
	$V_i T_i^c = V_f T_f^c$		2E.2b
Adiabats	$p_f V_f^\gamma = p_i V_i^\gamma$		2E.3
	$\gamma = C_{p,m} / C_{V,m}$		

FOCUS 2 · RECAP

Summary of the First Law

热力学第一定律总结

1. Energy and internal energy · 能量与内能
2. Enthalpy and thermochemistry · 焓与热化学
3. State functions · 状态函数
4. Adiabatic changes · 绝热变化

The First Law at a glance

Key concepts · 核心概念

- ΔU = q + w; energy conserved. · 第一定律
- H = U + pV; ΔH = q_p; Hess and Kirchhoff. · 焓
- U and H are state functions; work and heat are not. · 状态函数
- Perfect gas: U = U(T), so ΔU = Cv ΔT.
- Adiabatic changes satisfy pV^{γ} = constant. · 绝热

These five results are the whole First Law of thermodynamics.

Discussion questions

- Why can we measure ΔU but never absolute U?
- Why is ΔH the natural quantity for constant-pressure reactions?
- How does Hess's law let us predict reactions we cannot run?
- Why does a gas cool on adiabatic expansion but not on isothermal expansion?

· Think about state functions versus path functions.

FOCUS 2 · NEXT

Looking ahead

- The First Law tracks energy — but not the direction of change.
- The Second Law (Focus 3) introduces entropy and Gibbs energy.
  熵与 Gibbs 能
- There you will learn why some spontaneous changes occur.
  自发性

· Energy is conserved; entropy decides direction.